

WUDAOLIANG, China

WMO: 529080

Lat: 35.2156N

Lon: 93.0806E

Elev: 15135

StdP: 8.25

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-10.5	-34.2	1.4	2.5	-30.9	1.7	4.7	40.4	17.4	37.6	19.0	5.7	250	0.941

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	60.3	42.8	57.2	41.7	54.5	40.6	46.5	55.5	45.0	53.0	43.4	50.9	10.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
43.0	73.4	50.3	41.5	69.3	48.6	40.1	65.5	47.0	24.1	55.7	23.0	53.3	21.9	50.9	51.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.4	28.7	24.3	-19.3	65.2	2.8	3.2	-21.3	67.5	-22.9	69.4	-24.5	71.1	-26.5	73.4		
(5)			-20.2	48.7	2.6	1.7	-22.1	49.9	-23.7	50.9	-25.1	51.9	-27.0	53.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.1	5.1	9.0	15.4	24.5	33.0	39.5	44.7	44.1	37.7	25.6	14.4	7.7
	DBStd	15.01	5.87	6.61	6.43	6.22	4.61	4.65	4.36	4.30	4.73	6.18	5.33	5.62
	HDD50	9089	1393	1147	1072	764	528	317	173	190	369	756	1068	1312
	HDD65	14549	1858	1567	1537	1214	993	766	630	649	819	1221	1518	1777
	CDD50	15	0	0	0	0	0	1	8	6	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	9.7	12.7	13.3	11.9	10.2	9.3	8.2	7.6	7.1	7.1	7.8	10.2	11.4
Precipitation	PrecAvg	10.4	0.0	0.1	0.1	0.3	0.8	1.8	2.8	2.5	1.7	0.3	0.0	0.0
	PrecMax	16.0	0.2	0.2	0.4	0.7	1.9	4.3	5.2	5.1	3.3	1.5	0.4	0.2
	PrecMin	5.4	0.0	0.0	0.0	0.0	0.1	0.3	1.0	0.6	0.1	0.0	0.0	0.0
	PrecStd	1.9	0.0	0.0	0.1	0.2	0.4	0.8	1.1	1.0	0.7	0.3	0.1	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.0	33.7	41.9	48.3	55.0	60.1	65.4	65.2	56.5	49.8	35.7	31.7
		MCWB	18.4	20.2	24.8	28.8	33.3	41.8	45.4	43.2	41.8	32.1	22.0	18.7
	2%	DB	23.8	29.2	36.4	44.4	50.5	55.5	60.7	60.4	52.6	43.1	31.6	26.5
		MCWB	14.0	17.5	22.1	27.0	32.1	39.8	43.5	42.8	39.9	29.7	19.8	15.7
	5%	DB	20.6	25.6	32.8	41.2	46.9	52.3	57.5	57.1	49.7	39.1	29.0	22.9
		MCWB	12.1	15.4	20.1	25.7	30.5	38.7	42.3	42.1	38.7	27.8	18.2	13.3
	10%	DB	17.2	22.4	29.7	37.9	43.6	49.2	54.6	53.9	46.8	35.7	26.2	19.6
		MCWB	9.9	13.6	18.4	24.3	29.6	37.6	41.3	41.4	37.0	25.9	16.7	11.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	20.5	25.3	30.8	36.9	46.5	49.1	47.9	44.8	34.7	22.6	19.1
		MCDB	28.5	32.3	40.2	43.0	45.5	53.8	59.3	56.5	52.4	43.2	34.3	31.1
	2%	WB	14.4	17.8	22.6	28.5	34.4	43.6	46.9	46.2	42.7	31.7	20.3	15.9
		MCDB	23.0	28.2	35.1	40.7	44.3	50.9	56.1	55.5	50.6	39.7	30.3	25.9
	5%	WB	12.3	15.8	20.7	26.6	32.7	41.4	45.2	44.9	40.9	29.5	18.7	13.7
		MCDB	20.2	24.7	31.8	38.9	42.2	48.5	53.4	53.0	47.1	36.8	27.9	22.4
	10%	WB	10.1	13.9	18.8	25.1	31.2	39.4	43.6	43.3	39.0	27.3	17.0	11.6
		MCDB	17.1	22.0	28.8	36.2	40.4	46.1	51.1	50.8	44.3	33.9	25.6	19.3
Mean Daily Temperature Range	5% DB	MDBR	22.8	23.5	25.1	22.7	18.7	16.8	17.3	17.6	16.3	17.7	22.7	23.2
		MCDBR	25.4	26.5	28.5	26.6	24.8	21.8	22.4	23.4	20.1	20.7	24.6	26.2
		MCWBR	18.2	17.9	17.8	13.4	10.3	9.3	8.7	9.1	9.1	9.7	15.4	18.2
	5% WB	MCDBR	24.9	25.1	27.3	23.7	19.2	17.7	16.8	17.4	16.8	17.7	23.2	25.5
		MCWBR	17.9	17.2	17.3	12.3	9.5	9.1	8.2	8.3	8.9	9.2	14.9	17.9
Clear-Sky Solar Irradiance	taub	0.174	0.198	0.246	0.286	0.307	0.302	0.295	0.287	0.258	0.224	0.184	0.169	
	taud	2.607	2.539	2.387	2.269	2.271	2.399	2.488	2.510	2.561	2.581	2.611	2.649	
	Ebn at Noon	346	343	329	317	309	309	311	313	320	327	338	344	
	Edh at Noon	22	27	35	42	43	38	34	33	30	26	22	20	
All-Sky Solar Radiation	RadAvg	1043	1299	1658	1970	2081	2029	2067	1905	1660	1430	1176	1014	
	RadStd	70	85	87	84	97	113	144	106	115	101	52	42	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.90	+1.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		+1.01	+2.02	N/A	N/A	+0.65	+0.47	-439	-457	N/A

Nomenclature: See separate page